

Microwave Control-Chain Spoofing of NV Magnetometers: Detection Limits and Reference-Sensor Design Rules

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Abstract—Diamond magnetometers based on nitrogen-vacancy centers are moving toward field applications such as magnetic anomaly navigation, where a wrong field estimate becomes a wrong position. The quantum sensor is hard to reach remotely; its classical microwave control chain is not. We study in simulation what a defender can detect and recover when an attacker manipulates that chain, and turn the results into hardware design rules. A drive-frequency offset below one part per million reproduces the photon statistics of a genuine field change for any Ramsey schedule, so the sensor cannot verify its own output, and secret interrogation timing stops only precomputed phase attacks. Falling back to an auxiliary reference sensor on alarm cannot push the rate of harmful outputs below the reference’s own error rate, and a closed-form design equation predicts a blind window of small spoofs that pass unnoticed, which sets how precise the reference must be. In a 30-pipeline simulation fixed before the run, a reference sharing the primary sensor’s clock releases a harmful estimate on almost every attack; post hoc, the design equation matches all eight reference conditions within one percentage point.

Index Terms—NV magnetometry, quantum sensor security, sensor spoofing, microwave control, reference sensor fusion, Ramsey magnetometry

I. INTRODUCTION

Nitrogen-vacancy (NV) magnetometers combine high sensitivity with room-temperature operation, which makes them candidates for field deployment [1]. One such application is magnetic anomaly navigation, where the measured field is matched against a map to obtain position [2]. In this setting a field error is a navigation error, so the integrity of the released estimate matters as much as its precision. The question is where an attacker could corrupt it.

The weak point is not the quantum spin but the electronics that control it. An NV Ramsey magnetometer reads the field as a phase that accumulates between two microwave pulses. That phase is measured against a reference set by classical hardware: a reference clock, a microwave synthesizer, an arbitrary waveform generator (AWG), and the cables between them. Signal injection into the analog front ends of sensors is well documented [3], [4]. The manipulation an NV sensor needs is also small. A 2.8 kHz drive offset, 0.89 ppm of the 3.15 GHz transition, imitates a 100 nT field change. Every tone derived from a shifted clock inherits the clock’s fractional error, so pulling the usual 10 MHz reference by about 10 Hz, or rewriting the synthesizer’s frequency setting, is enough. This raises a first question: can the sensor notice?

Existing defenses cannot answer this, because none was designed for an attacker on the drive. NV sensing research treats control errors as benign imperfections to be averaged out by robust or double-quantum sequences [5]–[7]. Spoofing defenses for classical active sensors hide probing times from the attacker [8], a strategy that fast attackers can bypass [9]. A common engineering answer is to fall back to a second sensor when the two disagree, and such detectors are judged by their alarm rate. For a field estimate, however, what matters is whether the released value is correct. We therefore ask:

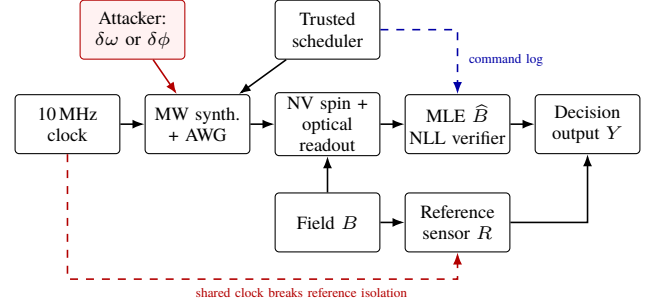


Fig. 1. Sensor chain and trust boundaries. The attacker offsets the drive frequency ($\delta\omega$) or phase ($\delta\phi$) after the trusted scheduler. The reference is secure only if it shares no clock, synthesizer or data path with the primary chain (red dashed).

when an attacker manipulates the microwave drive, what can the defender detect, how much can a reference sensor recover, and which hardware choices decide the outcome? We make three contributions:

- **Control-chain attack surface** (Section III). We map the standard detuning-field equivalence onto the control chain: a sub-ppm drive offset is invisible, secret timing stops only precomputed phase attacks, and dual-transition driving is a candidate defense.
- **Reference architecture and design equation** (Section IV). Fallback cannot beat the reference’s own error rate under attack, and a closed-form equation predicts residual harm and sizes the reference.
- **Engineering evaluation and design rules** (Sections V–VI). A simulation fixed in advance confirms the capability and floor results, a post-hoc comparison supports the design equation, and each countermeasure is mapped to its hardware. All results are simulated, not validated hardware performance.

II. SENSOR CHAIN AND THREAT MODEL

The attacker’s lever is the part of the chain that turns commands into microwave pulses (Fig. 1). A trusted scheduler chooses the interrogation time t_k and analysis phase ϕ_k of each setting k and logs them. The synthesizer and AWG drive one NV transition, and the estimator computes a field estimate $\hat{B}$ from the photon counts and the logged commands. A verifier scores settings held out from the fit, and an optional reference sensor supplies an independent reading R of the same field. The decision stage combines them into the released output Y .

The attacker acts between the scheduler and the spin. It can add a constant offset to the drive frequency, or change the analysis phase of each pulse, either from a table prepared in advance or with real-time knowledge of the schedule. It cannot observe the true field or the

TABLE I
MEASURES AND SIMULATION SETTINGS OF THE FROZEN RUN.

Harm	Fraction of attempts with released $ Y - B > \epsilon_H = 25 \text{ nT}$
Coverage	Fraction of attempts that release an output
RMSE	Root-mean-square error of released outputs
Alarm budget α	Fraction of clean sequences allowed to raise an alarm; 1%
Ramsey model	Eq. (1): $T_2^* = 100 \mu\text{s}$, $\eta = 0.1$, $c = 1$; phases $0, \pi/2$
Schedules	Fixed $t \in \{5, 10, 20, 40, 60\} \mu\text{s}$, or secret $t \sim U[5, 60] \mu\text{s}$ per setting
Sequence	20 settings $\times$ 1000 shots (half fit, half verify); $B \sim U[0, 1.9] \mu\text{T}$
Calibration	4,096 clean sequences per pipeline; thresholds are empirical quantiles
Intervals	95% bootstrap over whole pipelines (10,000 draws; 2,000 post hoc)
Attacker table	Differential evolution, 4 starts $\times$ 30 members $\times$ 24 generations; 64/128 training/validation sequences; offsets in $\pm\pi$
Spin-1 check	NV $S=1$, ^{14}N $I=1$; hyperfine $-2.14/-2.70 \text{ MHz}$, quadrupole -4.95 MHz ; 25 ns pulses, 10 MHz Rabi, no RWA; $B_0 = 5, 10, 20 \text{ mT}$

reference, and it cannot alter counts, logs, software or the reference path. Such access would bypass any measurement check and calls for data-path protection instead.

We judge a defense by what it releases, not by how often it alarms. Table I defines the measures. A release is *harmful* if it misses the true field by more than 25 nT, while the attacker aims to shift the estimate by 100 nT without an alarm; both values are evaluation parameters, not operational standards. Because a defense can avoid harm by releasing nothing, we also report coverage and the error of released outputs. With these measures, we can ask what the primary sensor can detect on its own.

III. CAN THE SENSOR DETECT THE ATTACK ON ITS OWN?

A Ramsey sequence turns the field into a phase and the phase into a spin population. The first microwave pulse creates a superposition, and the spin then precesses for a time t at a rate set by its detuning from the drive. The second pulse, with analysis phase ϕ , converts the accumulated phase into a population that the optical readout measures. The probability of reading the $m_s = 0$ state is [10], [11]

$$p = \frac{1}{2} [1 + (1 - 2\eta) c e^{-t/T_2^*} \cos \Theta], \quad \Theta = (\gamma B + \delta\omega)t + \beta - \phi, \quad (1)$$

where γ is the gyromagnetic ratio, B the field and $\delta\omega$ any detuning of the drive. The preparation phase is β , the contrast c , the dephasing time T_2^* , and η a symmetric readout error. Repeating each setting N times gives binomially distributed photon counts.

Frequency offsets are undetectable. The physics here is standard; its consequence for the control chain is the point. Equation (1) depends on the field and the drive only through their sum $\gamma B + \delta\omega$. An attacker who lowers the drive frequency by $\gamma\Delta B$ therefore produces exactly the count distribution of a field $B + \Delta B$. This holds for any pulse shape and any schedule, including secret ones, as long as the dynamics of the addressed transition depend on the field only through this detuning. Because the clean and attacked data have the same distribution, every test computed from the primary sensor's data raises an alarm with the same probability in both cases. In the terms of cyber-physical security, the attack is *undetectable*: its measurements are exactly those of a legitimate state [12].

The equivalence is exact only in the two-level model, so we checked the neglected physics. Equation (1) ignores the second, unaddressed NV transition, the counter-rotating part of the drive, and the ^{14}N nuclear spin. We simulated the full coupled electron-nuclear system (Table I) at eight random settings per bias field. The largest difference between the populations under a real field change and under the equivalent drive offset was 1.1×10^{-4} at a 10 mT bias field and 4.8×10^{-4} at 5 mT. For comparison, the spoof itself changes the population by a median of 0.034, and shot noise at 1000 shots is 0.016 per setting. If the residual stays at this level, even an ideal detector that knew it exactly would need at least 5,493 sequences at 10 mT (293 at 5 mT) to detect it with 50% probability at a 1% false-alarm rate. The attack is therefore undetectable within any single sequence. Can the defender do better by hiding the schedule?

Secret timing stops only precomputed phase attacks. A phase actuator imitates a field change only if it applies $\delta\phi_k = -\gamma\Delta B t_k$ with the true interrogation time. If the defender draws each t_k secretly, an attacker that committed to a table built for public nominal times t_{nom} leaves a residual phase $\gamma\Delta B (t_{\text{nom}} - t_k)$ that varies from setting to setting. No single field value then explains all settings, and a likelihood test on held-out settings detects the misfit. A frequency offset needs no knowledge of t_k , because the extra phase grows with time automatically. Secret timing therefore does not help against it.

Dual-transition driving cancels common clock errors. The NV ground state offers a hardware defense. It has two microwave transitions, $m_s = 0 \rightarrow +1$ and $m_s = 0 \rightarrow -1$, whose frequencies move in opposite directions when the field changes: $\omega_{\pm} = D \pm \gamma(B_0 + B)$, where $D \approx 2.87 \text{ GHz}$ is the zero-field splitting and B_0 the bias field. A clock error, by contrast, shifts both drive tones in the same direction. Differencing the two measured detunings keeps $2\gamma(B_0 + B)$ and cancels offsets common to both tones, including drifts of D [6], [7], [13]. A fractional clock error ϵ scales both tones. At $\epsilon = 1 \text{ ppm}$ and $B_0 = 10 \text{ mT}$ it biases a single-transition estimate by about 112 nT but a dual-transition estimate by only 10 nT, forcing the attacker to inject a precise differential offset on two tones. This is an algebraic bound, not a demonstrated defense. A single-transition sensor therefore needs help from outside the chain; how much can an external reference provide?

IV. CAN A REFERENCE SENSOR RESTORE THE ESTIMATE?

Two checks raise the alarm. Each sequence splits its settings into an estimation set, on which the maximum-likelihood estimate $\hat{B}$ is fitted, and a verification set. The first check, S_N , is the negative log-likelihood of the verification counts under $\hat{B}$; it grows when the data do not fit a single field. The second check, $S_R = |\hat{B} - R|$, compares the primary estimate with the reference reading. The defender first fixes a false-alarm budget α , the fraction of clean sequences allowed to raise an alarm; we use $\alpha = 1\%$. Each check receives half of the budget: its threshold is set so that only a fraction $\alpha/2$ of its scores on clean calibration sequences lie above it [14]. The alarm fires when either check exceeds its threshold, so on clean data it fires on at most α of sequences. We call the discrepancy threshold κ .

The reference is only useful if it is isolated. Ideally, the reference measures the same field through its own electronics, so the attack cannot reach it. In practice, it may share a clock or synthesizer with the primary sensor, or carry an uncorrected calibration offset. This matters because whatever the reference shares with the primary chain, the attacker can shift too. We model the reading as

$$R = B + b_R + \lambda\Delta B + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma_R^2), \quad (2)$$

where σ_R is the reference noise, b_R a calibration offset, and λ the fraction of the spoof ΔB that also reaches the reference. An independent reference has $b_R = \lambda = 0$. A second NV sensor driven from the same clock has $\lambda = 1$.

Output policies. Once the alarm is decided, the decision stage chooses what to release. *Fallback* releases R on alarm and $\hat{B}$ otherwise. *Abstention* releases $\hat{B}$ only without an alarm and releases nothing otherwise. *Always-reference* ignores the primary sensor. *Fusion* rules blend the two sensors instead of switching. Inverse-variance fusion releases $R + w(\hat{B} - R)$ with $w = \sigma_R^2 / (\sigma_R^2 + \sigma_P^2)$, weighting each sensor by its precision, where σ_P is the primary sensor's clean error. Gated fusion uses fallback on alarm and inverse-variance fusion otherwise. Clipped fusion releases $R + \text{clip}(\hat{B} - R, \pm\kappa)$, so it follows the primary sensor only within the alarm threshold. Which of these policies limits harm under attack?

The reference sets a floor. A fallback release is harmless only if the released sensor is accurate. When the spoof makes $\hat{B}$ harmful, only a released, accurate R can avoid harm, so fallback harm is at least the reference's own harm rate, whatever the alarm rule. For an independent reference with Gaussian noise, that rate is $2\Phi(-\epsilon_H/\sigma_R)$, where Φ is the standard normal cumulative distribution function. This is the probability that the reference alone misses the true field by more than ϵ_H . The same floor holds in the worst case for every rule of the form $Y = R + g(\hat{B} - R)$ with any function g . This class includes fallback, all fusion rules above, and any alarm on $\hat{B} - R$. It assumes that the attacker chooses the spoof size without bound and that the reference error is Gaussian and independent of the spoof. It also assumes that the primary error depends on neither the field nor the spoof, and that the defender uses no prior on the field. Under these conditions $\hat{B}$ carries no usable information about B . Rules that exploit a bounded field prior are outside this statement. A floor is only a lower bound, though. How close does fallback get to it?

Design equation. Assume the spoof shifts the primary estimate by exactly $d > h$, with $h = \epsilon_H$, and neglect the primary sensor's own noise, which is small next to the reference's. The reference then reads $R = B + s + \epsilon$ with shift $s = b_R + \lambda d$. Harm arises in two disjoint ways, which give the two terms of

$$H_F(d) = P(|s + \epsilon| > h) + P(|s + \epsilon| \leq h, |d - s - \epsilon| \leq \kappa). \quad (3)$$

The first term covers an inaccurate reference: the released value is harmful whether or not the alarm fires, because both sensors are wrong. The second term covers an accurate reference that still leaves the discrepancy $|\hat{B} - R|$ below κ . No alarm fires, so the spoofed $\hat{B}$ is released.

For an independent reference ($s = 0$), the discrepancy is close to d , and the second term separates three regimes of spoof size:

- *Blind window*, $h < d \leq \kappa - h$. The spoof is large enough to be harmful but too small to push an accurate reference past the threshold. Every accurate reference is silent, so harm is 1.
- *Transition region*, $\kappa - h < d < \kappa + h$. The alarm fires for some reference noise values and not others, so harm falls as d grows. This region matters because moderate spoofs land here and are only partly caught.
- *Floor*, $d \geq \kappa + h$. Every accurate reference raises the alarm, so only the first term remains.

Because the discrepancy can deviate in either direction, its check is two-sided: the budget $\alpha/2$ is split equally between large positive and large negative discrepancies. This gives

$$\kappa \approx z_{1-\alpha/4} \sqrt{\sigma_R^2 + \sigma_P^2}, \quad (4)$$

where $z_{1-\alpha/4} = 2.807$ is the standard normal quantile. The blind window exists whenever $\kappa > 2h$. With the simulated primary noise $\sigma_P \approx 4.28$ nT, this happens once $\sigma_R > 17.3$ nT. At $\sigma_R = 20$ nT, spoofs from 25 to 32.6 nT pass unnoticed. Equations (3) and (4) thus become a sizing rule for the reference. The next section tests them.

V. EVALUATION

Data sources. Every result comes from a single simulation run whose cases, policies, endpoints and random seeds were fixed and time-stamped before it ran, which we call the frozen run. Two analyses added afterwards are marked post hoc: the design-equation comparison (Fig. 2) and the fusion replay (Table IV), which re-evaluates output rules on the stored records without new simulation. The spin-1 simulation and clock arithmetic of Sections I–III are separate calculations.

Setup. Each pipeline calibrates its thresholds on its own clean sequences (settings in Table I), taking the empirical quantile that leaves $\alpha/2$ of scores above it, and fits its own attacker table. The defender uses a fixed public schedule, the same schedule with random analysis phases, or secret times; each is a separate case, so the benefit of secrecy can be isolated. The run contains 20 cases. Attack cases vary the actuator (phase or frequency), the schedule, and the attacker's knowledge of the actual times. Reference-stress cases vary $\sigma_R \in \{5, 15, 50\}$ nT, add a 10 nT offset or a shared fraction $\lambda \in \{0.5, 1\}$, or shrink the spoof to 50 nT. Control cases include no attack, a genuine 100 nT field change, and uncalibrated readout drift. Thirty independently seeded pipelines, each with 512 clean/attacked pairs per case, give 15,360 attempts per case and condition. Unless stated otherwise, results refer to the *nominal configuration*: a 100 nT frequency spoof, $\sigma_R = 20$ nT, and thresholds calibrated at $\alpha = 1\%$.

Secret timing stops phase tables but not frequency offsets. Table II reports how often each attack reaches its 100 nT target without an alarm, using only the likelihood check. In the disclosure rows, the attacker learns each actual time either before the $2\mu\text{s}$ its actuator needs to apply the phase (timely) or after it (late); full knowledge gives the time immediately. Mean-time and zero-offset tables are simple baselines for the optimized table. A nominal-time phase attack succeeds on 14,758 of 15,360 attempts against the fixed schedule but on only 1 against secret times. Randomizing the analysis phase alone does not help, because the attacker simply adds its offset to whatever phase is used. The attacker's strongest option against secret times is a precommitted phase table: one phase offset per public nominal-time slot, chosen by numerical optimization (differential evolution) on simulated data to shift the estimate while still fitting the verifier. It shifts the estimate in 6,081 attempts but passes the verifier in only 17; since the search is finite, this is a lower bound on what a phase attacker can do. A drive-frequency offset needs no schedule knowledge and succeeds on 14,892 of 15,360 attempts even against secret times, with simulated count distributions matching the field-shifted ones to within 5.55×10^{-16} . With the primary sensor blind to this attack, everything rests on the reference.

Fallback reaches the floor, and the design equation explains it. In the nominal configuration, fallback releases a harmful estimate on 20.78% [20.26, 21.30] of attacks, the same rate as always-reference. This is the reference's own error rate at $\sigma_R = 20$ nT, so no alarm rule can remove it. In a post-hoc comparison, the design equation reproduces this and every other tested condition without fitting to attack data. Evaluated with the calibrated thresholds, it agrees with the observations within 0.60 percentage points in all 8 reference conditions of the setup (Fig. 2). The equation also explains the failures. A 50 nT spoof falls in the transition region and raises harm

TABLE II

ATTACK CAPABILITY, LIKELIHOOD CHECK ONLY (FROZEN RUN, 15,360 ATTACKS PER ROW). TARGET: 100 nT SHIFT REACHED; UNDETECTED: WITHOUT ALARM. DETUNING: FREQUENCY OFFSET.

Attack / defender schedule	Target	Undetected
Nominal phase / fixed	14,879	14,758
Nominal phase / hidden phase	14,669	14,593
Nominal phase / hidden time	2,569	1
Late disclosure / hidden time	2,569	1
Timely disclosure / hidden time	15,004	14,892
Full knowledge / hidden time	15,004	14,892
Detuning / hidden time	15,004	14,892
Selected phase table / hidden time	6,081	17
Mean-time phase / hidden time	5,900	14
Zero offset / hidden time	0	0

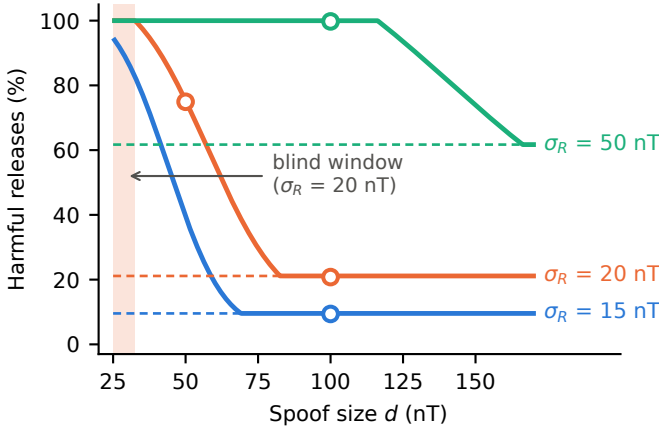


Fig. 2. Fallback harm from Eq. (3) (lines, post hoc), reference floors (dashed), and frozen-run observations with 95% intervals (circles). Shading: blind window at $\sigma_R = 20$ nT, spoofs of 25–32.6 nT that are harmful but raise no alarm.

to 74.93%. A noisier reference with $\sigma_R = 50$ nT widens κ until even the 100 nT spoof sits in the blind window, and fallback harm rises to 99.76%, worse than the reference alone (62.25%). A precise reference with $\sigma_R = 5$ nT removes harm entirely. Reference precision is one hardware requirement; isolation is the other.

A shared clock defeats the reference. Every departure from isolation in Eq. (2) raises harm (Table III). A reference that inherits the full spoof, as a second NV sensor on the same clock would, releases harmful estimates on 99.98% of attacks, yet alarms on only 0.94%, because both sensors move together.

What fallback, abstention and fusion buy. If fallback cannot beat the reference under attack, why use the primary sensor at all? The answer is clean precision. Without attack, fallback releases the primary estimate most of the time, and its RMSE is 6.14 nT, against 19.98 nT for always-reference. The paired difference is -13.84 nT, with an interval of [-14.27, -13.44] adjusted for multiple comparisons. This precision costs 20,000 primary shots per estimate, about 0.65 s of free precession with secret times before initialization and readout overheads. Reference integration time and cost depend on the platform and are not modeled. The fusion replay in Table IV shows the price of pushing precision further. Inverse-variance fusion is the most precise rule on clean data, but it trusts the precise, compromised primary sensor and releases a harmful estimate on every attack. Gated and clipped fusion fall in between, and every rule except always-reference has a worst case far above the reference floor, which is only a lower

TABLE III

REFERENCE DEPENDENCY AUDIT (FROZEN RUN): HARMFUL RELEASES OF FALLBACK UNDER A 100 nT SPOOF, $\sigma_R = 20$ nT.

Shared dependency	Model effect	Harm
None (independent)	$b_R = \lambda = 0$	20.78%
Calibration or offset error	$b_R = 10$ nT	26.94%
Partially coupled paths	$\lambda = 0.5$	89.67%
Shared clock or synthesizer	$\lambda = 1$	99.98%
Shared data path/software	attacker rewrites R	out of scope

TABLE IV

POST-HOC REPLAY OF OUTPUT RULES (SECTION IV) ON FROZEN RECORDS, $\sigma_R = 20$ nT. WORST CASE: MODEL MAXIMUM OVER 0–300 nT SPOOFS.

Rule	Clean RMSE (nT)	Attacked harm (%) 100 nT	Worst case (%) 50 nT	Worst case (%)
Always reference	19.98	20.78	20.78	21.2
Combined fallback	6.14	20.78	74.93	96.1
Gated fusion	6.09	20.78	74.93	95.4
Inverse-variance fusion	4.29	100.00	100.00	100.0
Clipped fusion	4.38	94.86	94.86	94.9

bound.

Abstention trades availability for harm: it lowers harm by 18.85 percentage points only by cutting coverage under attack by 98.07 points. All 296 outputs it still releases are harmful, and each release costs 1.04 million primary shots. Without attack, the alarm fires on 1.07% [0.91, 1.24] of sequences, and a genuine 100 nT field change raises only 154 alarms, because both sensors see it.

Averaging the reference is a cheaper route to precision if its errors are independent: 11 readings give $\sigma_R = 6.03$ nT, matching fallback's clean precision with a floor of 0.003%. Fallback pays off only if primary shots are far cheaper or drift prevents averaging.

VI. DESIGN RULES AND LIMITATIONS

The results reduce to four design rules, each with its hardware cost:

- *Match the defense to the actuator.* Secret timing needs only a randomized scheduler; it stops precomputed phase attacks, not frequency offsets.
- *Size the reference to the tolerance.* A 25 nT tolerance needs $\sigma_R < 17.3$ nT of total error, including drift and alignment.
- *Audit shared hardware.* The reference needs its own clock, synthesizer and data path; sharing a clock gives 99.98% harm.
- *Consider a dual-transition drive.* A second tone (synthesizer channel or IQ sideband) should cancel common clock errors (112 to 10 nT per ppm, analytic); it is untested against two-tone attacks.

These rules rest on simulation, not hardware. The Gaussian reference omits drift, bandwidth, alignment, spatial mismatch and correlated noise, so a real reference must be characterized first. The attacker is an ideal offset, not a circuit-level clock; its phase-table search is finite; the pipelines share one code base; and full proofs of the floor exceed this paper.

VII. CONCLUSION

In simulation, an NV Ramsey magnetometer cannot tell a sub-ppm drive offset from a field change. A reference restores integrity only down to its own error rate, only if precise enough to close the blind window, and only if it shares no clock with the sensor. Next come circuit-level clock models, two-tone attacks, and NV hardware.

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